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(Exaggerated royalty-free stock image of the common portrayal of a computer scientist. Will remove if images are not supposed to be here)

In popular culture, the computer scientist is often portrayed alone in a dark room typing away at a computer console. Within the actual discipline, this often can't be further from the truth. Computer Science is more than just the study of machines and computers, it is a community of practice where writing functions as the connections between humans and computers. Whether through the syntax of a Java file or the words in a peer-reviewed paper, writing in CS is a dynamic, multimodal act that establishes professional legitimacy and mediates the progress of science itself.

As I explored through interviews with an instructor and a peer, the transition into this field involves a shift in identity. One must move away from a "problem-solving" mindset learned in school (what my instructor interviewee called the "gold star" approach, for the “gold star” stickers teachers may give out) toward a communication mindset, where the value of a solution is determined by how effectively it can be understood, critiqued, and built upon by others. By examining these perspectives alongside threshold concepts, this paper will illustrate how writing in Computer Science

reflects shared values and supports the collective participation necessary for the field to evolve.

If writing is a “dynamic, multimodal technology for making meaning,” as the threshold concept suggests, then Computer Science is perhaps the ultimate example of this definition. However, what actually counts as “writing” depends on who you ask within the community. For the peer I interviewed, the definition is broad and functional: "code is definitely writing to me." Through this view, Python or C++ are simply languages with a unique audience: the computer itself. My instructor, however, observed a divergence within the field’s genres. He noted that while a software package or block of code provides the functional implementation of a solution, the academic paper is a separate layer that links to the code rather than being a translation of it. This is an important note that suggests writing in CS is not a single act, but a multimodal stack. At the bottom layer, code mediates the activity between programming and machine, while at the top, scholarly papers mediate the activity between researchers and the greater scientific community. To be a practitioner of Computer Science is to be a translator between all of these modes, ensuring both that the machine executes the logic and the human community understands its significance.

While the medium of writing may shift, the underlying values of the discipline seem to be focused on precision and transparency. In Computer Science, these are more than just stylistic preferences, and are instead fully functional. My peer interviewee emphasized that in code, precision is vital. If the code isn’t logically sound, the “audience” (computer) will fail to respond in the way the programmer is looking for. However, this value extends to the social interactions of the field as well. My peer talked

about their time working at an internship and noted that comments on his pull requests (essentially formal requests to merge new code into a main project) initially felt angry or personal. He soon realized that the bluntness was a reflection of the community's values. The goal is “functioning code,” over the protection of the writer's ego. This aligns with Carolyn Miller's argument that genre is "centered not on the substance or the form of discourse but on the action it is used to accomplish" (Miller 151). In this case, the genre of the "Code Review" serves the social action of maintaining/expanding system functionality, which encourages a transparent, egoless communication style.

The transition from a student to a professional in Computer Science is marked by a shift in how one defines 'success' in writing. My instructor described this as moving away from the 'gold star' approach of k-12 and early college, where the goal is simply to solve the problem to get credit. This mirrors the findings of Johnson and Krase, who observe that the graduating seniors often point to “the importance of junior- and senior-level tasks to their development as writers” (2). They suggest that for a student to truly "come to learn" as a writer, they must move beyond a "paper-for-grade economy" (10). As my instructor noted, "you can solve a problem, but it won't matter if you can't communicate that you have solved it." This realization marks the point where a student stops "simplifying a writing task" to make it accessible for a grade and begins to realize their own ideas as part of a "coherent field of diverse voices" (Johnson and Krase 2, 14). This illustrates the threshold concept that writing enacts disciplinarity. The act of writing the 'story' of the code is the act of becoming a scientist. By moving away from the 'gold star' mindset, the student begins to use writing to expand professional disciplinary knowledge rather than just satisfying a rubric. Writing in a discipline is a way

of “coming to learn.” The writing does not just report the result, but also constructs the significance of the result. Writing moves from a passive “I found this” to an active, often persuasive argument for why the discovery matters to the wider scientific ecosystem.

If writing in Computer Science includes writing the “story” of a solution, then that story is frequently designed to provoke a specific response. This is related to the threshold concept that writing mediates activity. My instructor interviewee noted that he enjoys a more “aggressive” writing style, especially when relating to a new software package. He describes this genre as often inherently persuasive. Writers are essentially saying “My package is better; here is why you should use it.” This is also an illustration of Miller’s “Genre as Social Action.” In this context, a software writeup or a research paper is more than a neutral technical description, it is a social tool used to convince a community to adopt a new algorithm, shift their methodology, or “act in some manner” to advance the field. Writing mediates the activity between the creator’s discovery and the community’s adoption. Without a rhetorical ‘push’, even the most useful code would remain stationary and never cross from being a private project to a public contribution.

Finally, writing serves as a representation of identity, where the quality of the writing or code directly constructs the author’s professional persona. In computer science, your writing is your proxy. It is the primary way you establish what my instructor calls “legitimacy and competence.” He pointed out that the consumer of your writing uses the clarity or precision of your writing to judge the quality of your mind. “Your writing defines you,” my instructor noted. This creates a high-pressure environment for the writer. My peer agreed with this sentiment, and admitted he felt a sense of anxiety when writing for a mathematical modeling class. He was worried that if he was

perceived as a “bad communicator,” then others would be led to believe they didn’t understand the underlying math, even if the calculations were perfect. This reveals something about the discipline: to be seen as a legitimate computer scientist, one must be able to write themselves into that role, proving that identity as a computer scientist is linked to the CS documents we produce.

It has become clear to me that Computer Science is not the silent, purely logical practice it is often perceived to be. Instead it is a community where writing in many forms, whether via code, documentation, or scientific papers, serves as the primary vehicle for participation and progress. Writing is the tool that transforms a solution into science.

This project has started to reshape my understanding of my own identity within the field. Up until now, I have also had the “gold star” approach where most of my code is meant to comply with a strict rubric and metrics for a grade. By better understanding the values and ideologies that govern computer science writing now, I am more prepared to enter into the discipline through writing. Ultimately, I have learned that to be a successful computer scientist, one must

master the ‘full stack’ of computer science writing as much as being a master of code syntax.

Works Cited

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